

C-PGS: Causal Prior-Guided Style-Aware Semi-Supervised Semantic Segmentation for Ancient Chinese Architecture

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Abstract

Semantic segmentation of ancient Chinese architecture (ACA) is fundamental to digital preservation, 3D reconstruction, and virtual restoration. Due to the high cost of annotation and the scarcity of labeled data, semi-supervised learning offers a practical solution for this task. However, in ACA scenes, large intra-class variation and high inter-class similarity make models prone to style-induced appearance confounding. Such confounding can be repeatedly amplified through pseudo-label generation and retraining, thereby degrading segmentation performance. Motivated by a causal analysis of style-induced bias propagation, we propose Causal Prior-Guided Semi-Supervised Semantic Segmentation (C-PGS), a style-aware framework for semi-supervised semantic segmentation of ACA. Specifically, C-PGS achieves style awareness by constructing a style hypergraph to capture style-related higher-order co-occurrence and exclusion relationships among architectural components, and uses the inferred style prior to regularize pseudo-label generation. In addition, C-PGS introduces SAM2-generated instance masks as geometric priors to improve boundary consistency and structural completeness for fine-grained components. Moreover, C-PGS is model-agnostic and can be readily integrated into multiple semi-supervised semantic segmentation baselines. Extensive experiments on the ACA dataset show that C-PGS consistently improves segmentation performance across five baseline frameworks under different labeling ratios.

Keywords: Semi-supervised semantic segmentation, Ancient Chinese architecture segmentation, Causal analysis, Hypergraph priors, Instance-level geometric priors

1. Introduction

Ancient Chinese architecture (ACA) is an important carrier of human history and culture, containing rich structural information, artistic value, and distinctive regional styles. Semantic segmentation of ACA [1] provides reliable component-level priors for downstream tasks such as 3D reconstruction [2], damage detection [3], and restoration planning [4]. However, compared with natural-scene imagery, ACA data are harder to collect and much more expensive to annotate. This challenge makes semi-supervised semantic segmentation particularly well-suited for ACA scenes, since it can effectively utilize limited labeled data together with abundant unlabeled samples. Existing methods mainly include pseudo-label-based approaches [5, 6] and consistency-regularization-based approaches [7, 8], both of which have shown strong performance in various vision tasks [9, 10]. Nevertheless, their effectiveness remains limited in ACA scenes.

The main challenge in semi-supervised ACA semantic segmentation arises not only from label scarcity but also from the intrinsic visual properties of the data, particularly large intra-class variation and high inter-class similarity. These properties make it difficult for models to rely on stable semantic cues and

may instead encourage them to exploit unstable appearance patterns under limited supervision. To provide empirical evidence for this ambiguity, Fig. 1 presents a unified analysis from feature construction to statistical and visual observations. Specifically, Fig. 1(a) shows how category-aware region embeddings are constructed from SAM2-based region features for feature-space analysis. Based on these embeddings, Fig. 1(b) compares the average intra-class distance d_{intra} with the nearest inter-class distance d_{nearest} for each category. The results show that d_{intra} exceeds d_{nearest} in 12 out of the 18 categories, indicating that many ACA components exhibit intra-class variation comparable to or even greater than their nearest inter-class separation. This fragility stems from two main factors: first, categories such as pillars exhibit substantial variations in material, texture, color, and geometry across architectural complexes (Fig. 1(c)); and second, some stone categories, such as stone lions, stone elephants, and stone tablets, share similar local textures and stone-pedestal structures, which may cause confusion in feature representation.

These characteristics make semi-supervised semantic segmentation particularly challenging in ACA scenes. The challenge is compounded by limited supervision and by models tendency to rely on style-induced appearance confounding associated with materials, textures, colors, and architectural styles. During semi-supervised training, such unstable correlations can be introduced into the teacher-generated pseudo-labeling pro-

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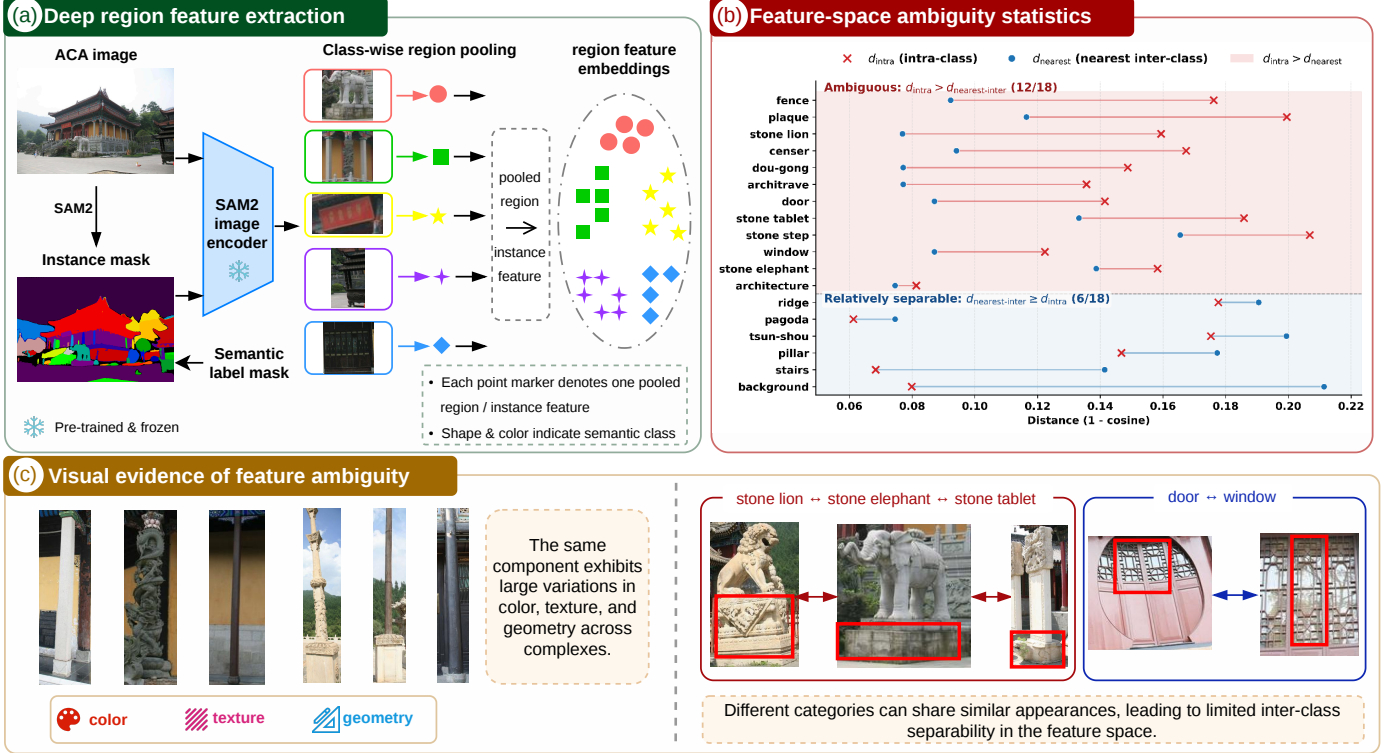


Figure 1: Statistical and visual evidence of feature-space ambiguity in ACA scenes. (a) SAM2 produces category-agnostic instance masks and deep feature maps. Since instance masks have no semantic labels, their categories are assigned according to overlap with ground-truth semantic masks, and the corresponding feature regions are pooled to obtain category-aware region embeddings. (b) We compare the average intra-class distance d_{intra} and the nearest inter-class distance $d_{nearest}$ for each category. (c) Visual examples illustrate large intra-class appearance variation and visually similar inter-class categories.

cess and gradually accumulate as style-induced pseudo-label bias propagation. Although recent semi-supervised semantic segmentation methods [7, 11] have been primarily developed and validated in natural-scene settings, they still lack explicit mechanisms to suppress this type of bias in ACA data. Recent studies [12, 13, 14] have shown that causal reasoning can help reduce reliance on spurious correlations and improve the robustness of semantic segmentation. This perspective is also relevant to ACA scenes, where style-induced appearance confounding can interfere with stable pseudo-label learning under limited supervision. Existing causal methods mainly address background bias, contextual dependence, or general feature confounding in natural scenes, while style-induced pseudo-label bias propagation in semi-supervised semantic segmentation of ACA remains largely unexplored.

In ACA scenes, style is reflected in appearance attributes such as materials, textures, colors, and decorative details, as well as in category co-occurrence, layout regularity, and local structural organization. Moreover, the same category may exhibit substantially different visual manifestations across architectural complexes, whereas some visually similar categories may still follow relatively stable co-occurrence or exclusion patterns under specific styles. These observations suggest that style jointly affects category distribution, visual manifestation, and pseudo-label reliability. Therefore, the central challenge in semi-supervised semantic segmentation of ACA lies in mitigating the continuous propagation of style-induced appear-

ance confounding under limited labeled data. To address this issue, we propose Causal Prior-Guided Semi-Supervised Semantic Segmentation (C-PGS), a style-aware framework for semi-supervised semantic segmentation of ACA motivated by a causal analysis of bias propagation. C-PGS incorporates two complementary priors. At the category level, we construct a style hypergraph based on higher-order co-occurrence and exclusion relationships among ACA components. This hypergraph encodes structured style constraints and is used to regulate pseudo-label generation and updating. At the instance level, we introduce instance masks generated by the vision foundation model SAM2 [15] as external geometric priors to improve local structural completeness and boundary consistency for fine-grained components. The main contributions of this work are summarized as follows:

- We reveal the dual role of architectural style in semi-supervised ACA segmentation from a causal perspective: it may induce appearance shortcuts that corrupt pseudo-label learning, while simultaneously providing structural cues to alleviate such bias. Based on this insight, we design plug-and-play prior modules that consistently improve generic segmentation models on ACA tasks.
- We propose a style hypergraph representation to model higher-order co-occurrence and exclusion relationships among architectural components, thereby improving

pseudo-label reliability and reducing inter-class confusion in ACA scenes.

- We introduce a SAM2-based instance-level geometric prior to enhance boundary consistency and structural completeness for fine-grained architectural components, reducing local omissions and fragmentation in segmentation results.
- Extensive experiments on the ACA dataset show that C-PGS is model-agnostic and can be readily integrated into five representative semi-supervised segmentation frameworks, consistently improving performance across different labeling ratios.

The remainder of this paper is organized as follows. Section 2 reviews the studies most relevant to this work, including semi-supervised semantic segmentation, causal inference, and Hypergraph-based Structured Representations. Section 3 presents the causal motivation and the SCM-based analysis of bias propagation in semi-supervised semantic segmentation of ACA. Section 4 describes the proposed C-PGS framework. Section 5 presents the experiments and results. Finally, Section 6 concludes the paper.

2. Related Work

2.1. Semi-Supervised Semantic Segmentation

Semi-supervised semantic segmentation aims to train segmentation models with a small set of pixel-level annotations and a large amount of unlabeled data. Existing methods mainly improve performance through pseudo-label learning, consistency regularization, or a combination of the two. Pseudo-label-based methods use high-confidence predictions on unlabeled images as supervisory signals, while consistency-based methods enforce stable predictions under different perturbations or views.

Among these directions, pseudo-label learning has become a central line of research. Existing methods improve pseudo-label utilization from several aspects, such as reliability modeling, feature augmentation, confidence calibration, and structural supervision. MMFA [16], for example, enhances the use of unlabeled samples through multi-reliability modeling and multi-level feature augmentation, while CSL [17] shows that high-confidence predictions are not always semantically correct in complex scenes. Other methods further improve unlabeled learning by introducing confidence-driven consistency [18], collaborative pseudo-label refinement [19], ranking-based supervision [20], feature perturbation [21], multi-constraint consistency [22], and cross-scale structural supervision [23].

Although these methods have substantially advanced semi-supervised segmentation, they are mostly developed and evaluated on natural-scene benchmarks, where category appearance is relatively stable and object boundaries are clearer. In ACA scenes, however, large intra-class variation and high inter-class similarity make models more likely to rely on unstable appearance cues. As a result, pseudo-label errors are harder to avoid

and more likely to be reinforced during teacher-student training. Existing methods rarely address this issue explicitly, and they do not directly consider how style-induced appearance bias propagates through the pseudo-label feedback loop.

2.2. Causal Inference in Semantic Segmentation

Causal inference has recently received growing attention in computer vision as a way to reduce spurious correlations and improve model robustness and generalization [24]. In semantic segmentation, causal modeling has been explored mainly for addressing background bias, contextual dependence, and feature confounding. Existing studies mainly follow two directions. One line of work uses causal intervention or deconfounding to suppress the influence of confounding factors and improve segmentation robustness. For example, CPCA [25] addresses performance degradation under domain shift in high-resolution remote sensing image segmentation through causal feature disentanglement and intervention learning. The other line focuses on counterfactual reasoning or invariant causal mechanism modeling. CausalCLIPSeg [26], for instance, introduces causal intervention into referring medical image segmentation to reduce confounding bias in vision-text alignment, while ICMSEG [27] improves the generalization of cross-modal medical image segmentation on unseen domains by modeling invariant causal mechanisms. CAUSE [28] further shows that causal modeling can also improve consistency and robustness in unsupervised semantic segmentation.

Although these studies highlight the value of causal reasoning for improving segmentation robustness and generalization, most existing work [29, 30] focuses on bias modeling in fully supervised or related settings. By contrast, limited attention has been paid to semi-supervised semantic segmentation, especially to how style-induced bias in pseudo-labels propagates and accumulates through the teacher-student loop.

2.3. Hypergraph-based Structured Representations

Conventional graph-based representations [31] have been widely used to model semantic relations in visual tasks. However, ordinary graphs mainly capture pairwise interactions, which limits their ability to represent higher-order structural dependencies among multiple categories. Hypergraph-based representations provide a more expressive alternative by allowing a single hyperedge to connect multiple nodes simultaneously. Recent studies have shown that hypergraph or hyper-relational representations are effective for modeling higher-order relations. For example, Vision HGNN [32] models images with hypergraph structures beyond pairwise relations, and Text2NKG [33] explores fine-grained n-ary relation construction for structured knowledge representation. Subsequent studies further improve these representations with neural or attention-based mechanisms [34, 35]. Nevertheless, most existing works mainly focus on general higher-order relation representation, embedding, or reasoning, while less attention has been paid to using hypergraph-structured relations as explicit structural priors for pseudo-label regulation in semi-supervised semantic segmentation. Different from previous work, our

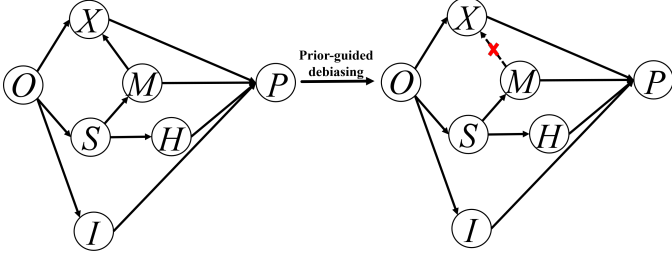


Figure 2: The structural causal model of the proposed framework for semi-supervised semantic segmentation in ACA scenes.

method uses a style hypergraph to encode higher-order co-occurrence and exclusion patterns among architectural components and employs it as a category-level structural prior to constrain pseudo-label generation.

3. Causal Motivation

3.1. Architectural Style as a Source of Bias and Constraint

Compared with natural scenes, ACA images usually exhibit larger intra-class variation and higher inter-class similarity. As shown in Fig. 1, the same semantic component may differ substantially in material-related appearance, texture, color, and geometric form across architectural complexes, while different categories may share similar appearances. Under limited supervision, these characteristics make it difficult for the model to establish stable semantic decision boundaries. In semi-supervised semantic segmentation, this issue is further aggravated because early prediction errors can enter the teacher-student pseudo-label loop and accumulate during self-training.

A distinctive property of ACA scenes is that architectural style plays a dual role. On the one hand, style can act as a source of appearance confounding. Variations in materials, textures, colors, and decorative patterns may encourage the model to associate semantic categories with superficial visual cues rather than stable semantic evidence. As a result, the teacher model may generate pseudo-labels that appear locally confident but are semantically unreliable, and such errors can be repeatedly reinforced during subsequent iterations. On the other hand, style is also reflected in relatively stable structural regularities, such as category co-occurrence, mutual exclusion, and part-whole compositional relations. These regularities provide useful constraints on which categories are likely to appear together, which are structurally incompatible, and how local components are organized within a complete architectural structure.

Therefore, the central challenge in semi-supervised segmentation of ACA is not merely generic pseudo-label noise, but the generation and accumulation of style-induced bias during pseudo-label learning. More importantly, the same scene style that introduces appearance confounding also contains structural information that can be exploited to suppress such bias. This observation motivates us to distinguish between two effects of style: unstable appearance correlations that may induce biased prediction, and stable structural regularities that can provide corrective guidance.

Table 1: Summary of the main notations used in C-PGS.

Symbol	Description
$\mathcal{D}_l$	Labeled training set.
$\mathcal{D}_u$	Unlabeled training set.
O	Input ACA image.
Y	Ground-truth segmentation mask.
$\hat{Y}_u$	Pseudo-label of an unlabeled image.
$\mathcal{C}$	Semantic category set.
C	Number of semantic categories.
S	Latent scene-level style factor.
M	Appearance-level style manifestation.
H	Structural style manifestation.
I	Instance-level geometric cue.
X	Feature representation extracted by the segmentation model.
$s_\theta(\cdot)$	Segmentation network with parameters θ .
Z_u	Teacher logits of an unlabeled image.
$\hat{Z}_u$	Style-prior-adjusted teacher logits.
π^{sp}	Category-level style prior.
π^{ip}	Instance-level geometric prior.
$g_H(\cdot)$	Style-prior inference function.
$g_I(\cdot)$	Instance-prior generation function.
$\mathcal{E}$	Reference hyperedge library.
e_j	Hyperedge of the j -th labeled image.
g_j, g_u	Architectural-complex identifiers of labeled and unlabeled images.
z_u	Category evidence vector of an unlabeled image.
P_u^+	Completed category set inferred from matched hyperedges.
$\mathcal{M}(O)$	Candidate instance mask set of image O .
m_k	The k -th candidate instance mask.
$\mathcal{T}(\cdot)$	Instance-guided refinement operator.
α	Strength of style-prior modulation.
λ_u	Weight of the unsupervised loss.
τ	Confidence threshold for pseudo-label filtering.
η	Dominance threshold in instance refinement.
r	Normalized refinement coefficient for mask expansion.
R_r	Pixel-level dilation radius converted from r .
$\mathcal{L}_{sup}$	Supervised loss.
$\mathcal{L}_{unsup}$	Unsupervised loss.

3.2. SCM-Based Analysis of Bias Propagation

Motivated by the above observation, we interpret semi-supervised semantic segmentation of ACA from a causal perspective and formulate a Structural Causal Model (SCM) [36], as shown in Fig. 2. Here, the SCM is not intended for strict causal effect identification. Instead, it serves as a conceptual tool for describing how style-related bias may arise and propagate in the pseudo-labeling process, and for motivating the design of the proposed structured priors.

Given an input image O , the segmentation network extracts class-discriminative features X and then produces the pixel-level prediction P . To describe the role of architectural style in this process, we introduce three style-related variables together with one external structural cue. Specifically, S denotes the latent scene-level style factor underlying a particular architectural complex. It is not a semantic category label, but refers to the shared appearance and structural characteristics of components within the same architectural complex. At the appearance level, style is reflected by materials, textures, colors, and decorative details; at the structural level, it is reflected by

category co-occurrence, mutual exclusion, and compositional organization. For example, components in the same complex may share similar stone materials and carving patterns, while the same semantic component may present different textures or geometric forms across different complexes. Accordingly, M represents the appearance-level manifestation of style, and H denotes the structural manifestation of style. In addition, we introduce I as an external instance-level geometric cue associated with shape integrity and boundary continuity. In practice, the latent factor S is not directly observable; instead, its influence is reflected through the observable appearance-related factor M and the structural factor H .

Based on these variables, the main dependencies are as follows. Observation of the image O provides visual evidence for feature extraction, and the learned representation X subsequently determines the prediction P . The latent style factor S influences prediction through two distinct pathways. The first is the appearance-related path $S \rightarrow M \rightarrow X \rightarrow P$, which captures how the model may rely on superficial cues such as materials, textures, and colors, thereby forming shortcut-based recognition patterns. Although such shortcuts may still produce high-confidence predictions, these predictions do not necessarily reflect stable semantic discrimination and can therefore become an important source of pseudo-label bias. The second is the structure-related path $S \rightarrow H \rightarrow P$, which reflects relatively stable regularities of category configuration under a given architectural style, including co-occurrence, exclusion, and compositional relations. Compared with appearance-level cues, this structural information acts more directly on category relationships and thus provides more reliable support for semantic prediction. In addition, the external geometric cue I provides complementary guidance through the path $I \rightarrow P$, which is particularly useful for correcting local structural errors such as omission, boundary discontinuity, and region fragmentation.

The SCM in Fig. 2 thus suggests two different information flows in semi-supervised segmentation of ACA: an appearance-driven bias flow that may distort pseudo-label learning, and a structure-guided constraint flow that can improve prediction reliability. This view suggests that the key is not to remove architectural style, but to reduce the unstable appearance correlations associated with it while retaining its more stable structural regularities. Since the latent style factor S is not directly observable and is not explicitly modeled, we translate this analysis into two forms of structured prior knowledge: a category-level structural prior for suppressing erroneous category propagation in pseudo-label generation, and an instance-level geometric prior for improving local structural completeness and boundary consistency. Section 4 instantiates these two priors as a hypergraph-based style prior for pseudo-label debiasing and a SAM2-based geometric prior for prediction refinement.

4. Method

Based on the analysis in Section 3, we introduce C-PGS as a style-aware framework for semi-supervised semantic segmentation of ACA. As shown in Fig. 3, the framework combines a category-level style prior π^{sp} with an instance-level geometric

prior π^{ip} under a teacher-student learning paradigm. The former regularizes pseudo-label generation by injecting category-level structural constraints into teacher predictions, which helps reduce erroneous category propagation caused by appearance-driven bias. The latter provides geometric guidance at inference time for refining local structures and correcting errors such as omission, boundary discontinuity, and region fragmentation. For clarity, the main notations used in C-PGS are summarized in Table 1.

4.1. Framework Overview

Given a labeled dataset $\mathcal{D}_l = \{(O_i, Y_i)\}_{i=1}^{N_l}$ and an unlabeled dataset $\mathcal{D}_u = \{O_j\}_{j=1}^{N_u}$, the segmentation network first extracts discriminative features X from an input image O and then produces pixel-level prediction logits $s_\theta(X)$. On this basis, we construct two complementary image-dependent priors:

$$\pi^{\text{sp}} = g_H(O), \quad \pi^{\text{ip}} = g_I(O), \quad (1)$$

where $g_H(\cdot)$ denotes the inference function of the category-level style prior, and $g_I(\cdot)$ denotes the generation function of the instance-level geometric prior. Specifically, π^{sp} is inferred from instance-level category evidence together with a style hypergraph constructed from labeled data, whereas π^{ip} provides local geometric constraints for structural refinement and is instantiated in practice from candidate instance masks generated by an external instance segmentation model.

During training, the category-level style prior π^{sp} is used to regulate the teacher predictions, yielding a more stable pseudo-label distribution:

$$P_s(Y | X, O) = \text{Softmax}(s_\theta(X) + \alpha \pi^{\text{sp}}), \quad (2)$$

where α is a scaling factor that controls the strength of prior modulation. The prior π^{sp} is broadcast over the spatial dimensions before being added to each class channel of the teacher logits. During inference, the instance-level geometric prior π^{ip} is introduced to refine the initial prediction:

$$\begin{aligned} \hat{Y}^{(0)}(x, y) &= \arg \max_{c \in C} P_s(Y(x, y) = c | X, O), \\ \hat{Y} &= \mathcal{T}(\hat{Y}^{(0)}, \pi^{\text{ip}}; \eta, r). \end{aligned} \quad (3)$$

where $\hat{Y}^{(0)}$ denotes the initial prediction mask obtained from the style-prior-adjusted prediction distribution $P_s(\cdot)$, $\mathcal{T}(\cdot)$ denotes the geometric-prior-guided refinement operator, η is the dominance threshold, and r is the normalized refinement coefficient for mask expansion.

Overall, the two priors play complementary roles in the proposed framework. The style prior π^{sp} improves the structural reliability of pseudo-labels during training, while the geometric prior π^{ip} enhances local completeness and boundary consistency during inference. Together, they improve segmentation quality in ACA scenes.

4.2. Style Prior for Pseudo-Label Debiasing

As discussed in Section 3, semi-supervised semantic segmentation in ACA scenes is challenged by limited pixel-level

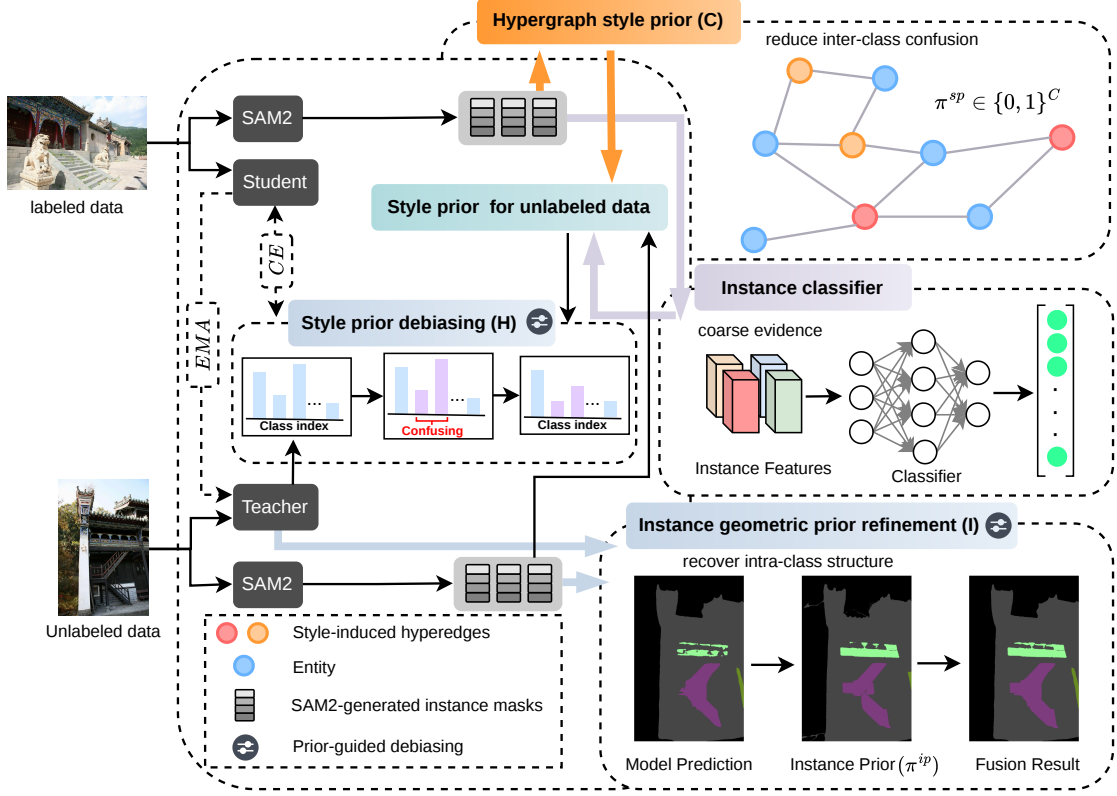


Figure 3: Overall framework of C-PGS for semi-supervised semantic segmentation of ancient Chinese architecture. The hypergraph-based style prior guides pseudo-label debiasing by modeling high-order category relations and suppressing structurally implausible categories. The SAM2-based instance geometric prior refines prediction masks by recovering local completeness and improving boundary consistency.

supervision and appearance-driven shortcuts. These shortcuts can be reinforced during pseudo-label self-training and gradually accumulate as category-level bias. To address this issue, we incorporate a category-level style prior π^{sp} into pseudo-label generation to guide teacher predictions with scene-level structural regularities. Specifically, the style prior is inferred in two stages. We first extract instance-level category evidence from the input image and then match this evidence against a style hypergraph constructed from labeled samples to infer the category-level prior for the current scene. The inferred prior is incorporated into the teacher logits as a category-level bias term, which suppresses category responses that are inconsistent with the structural pattern of the scene and reduces the propagation of biased pseudo-labels.

4.2.1. Instance-level Category Evidence Extraction

To infer the style prior for the current image, we first construct an instance-level category evidence extraction branch that provides auxiliary semantic cues. Given an input image O , we use SAM2 to generate a set of candidate instance masks,

$$\mathcal{M}(O) = \{m_1, m_2, \dots, m_K\}, \quad (4)$$

where K is the number of candidate instances and m_k denotes the k -th candidate mask. For each candidate region, the corresponding masked feature is extracted and fed into a lightweight

MLP classifier $f_{cls}(\cdot)$ to predict its semantic category:

$$\hat{c}_k = \arg \max f_{cls}(\Phi(m_k)), \quad k = 1, 2, \dots, K, \quad (5)$$

where $\Phi(m_k)$ denotes the feature representation of the k -th instance. The classifier is trained only on labeled images, and each candidate mask is assigned a category label based on its overlap with the ground-truth pixel annotations.

The predicted categories of all candidate instances are then aggregated into a binary vector

$$\mathbf{z} = [z_1, z_2, \dots, z_C]^T \in \{0, 1\}^C, \quad (6)$$

where C is the number of semantic categories. The resulting vector $\mathbf{z}$ represents the instance-level category evidence of the current image and is subsequently used as the query evidence for style-prior inference.

4.2.2. Hypergraph-based Style Prior Inference

Simple pairwise category co-occurrence is often insufficient to characterize architectural style, which is more naturally reflected by stable higher-order compositional relationships among multiple semantic categories. To capture such structural regularities, we introduce a hypergraph-based inference scheme that models higher-order category dependencies in ACA scenes and derives a category-level style prior π^{sp} . This design is motivated by the observation that recurrent higher-order category compositions provide informative cues for scene

style. In this work, style awareness refers to exploiting style-related structural regularities for pseudo-label regulation, rather than explicitly classifying architectural styles. The construction of the style hypergraph is illustrated in the upper part of Fig. 4.

Formally, let the labeled dataset be defined as

$$\mathcal{D}_l = \{(O_j, C_j, g_j)\}_{j=1}^J, \quad (7)$$

where O_j denotes the j -th labeled image, $C_j \subseteq C$ denotes the set of categories present in that image, and g_j denotes the architectural-complex identifier, referred to as building-group knowledge (BGK) in this work. It is used as image-level collection metadata and does not provide pixel-level or category-level semantic annotations. Based on the labeled data, we construct a reference hyperedge library

$$\mathcal{E} = \{e_j\}_{j=1}^J, \quad (8)$$

where each hyperedge is defined as

$$e_j = C_j \subseteq C. \quad (9)$$

Accordingly, semantic categories are treated as nodes, while each hyperedge corresponds to a higher-order category co-occurrence pattern derived from a labeled image. For an unlabeled image, its category evidence vector $\mathbf{z}_u$ is used to match the current scene against the reference hyperedge library. The style-prior inference process is illustrated in the lower part of Fig. 4.

For an unlabeled image O_u , we denote its architectural-complex identifier as g_u , which follows the same BGK definition as g_j . Based on g_u , we first retrieve the subset of reference hyperedges from the same architectural complex. We then match the instance-level category evidence $\mathbf{z}_u$ with these candidate hyperedges according to their category overlap. The matched hyperedges are further used for category completion, yielding the prior category set P_u^+ . The detailed matching and completion procedure is given in Algorithm 1. The resulting category-level style prior is represented as a binary mask $\pi_u^{\text{sp}} \in \{0, 1\}^{|C|}$, where

$$\pi_{u,c}^{\text{sp}} = \begin{cases} 1, & c \in P_u^+, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

The inferred style prior is not used to directly assign pixel labels. Instead, it acts as a category-level structural constraint on the teacher prediction for the unlabeled image. Let $Z_u \in \mathbb{R}^{H \times W \times C}$ denote the original teacher logits for O_u . We incorporate the style prior by modulating the logits as

$$\hat{Z}_u(x, y, c) = Z_u(x, y, c) + \alpha \pi_{u,c}^{\text{sp}}, \quad (11)$$

where α is a modulation coefficient controlling the strength of prior injection. The pseudo-label distribution is then obtained by

$$P_s(Y_u(x, y) = c \mid O_u) = \frac{\exp(\hat{Z}_u(x, y, c))}{\sum_{c'=1}^C \exp(\hat{Z}_u(x, y, c'))}. \quad (12)$$

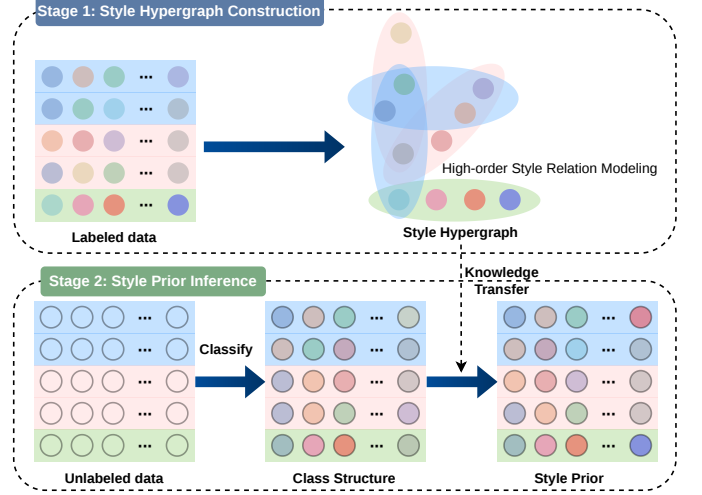


Figure 4: Two-stage process of style prior modeling. First, labeled samples are used to build a style hypergraph for modeling high-order category relations. Then, unlabeled samples are matched with the learned hypergraph to infer style priors that guide pseudo-label debiasing.

By introducing the hypergraph-based style prior into pseudo-label generation, the model is encouraged to preserve category combinations that are structurally consistent with the current scene, while suppressing implausible category responses that violate learned co-occurrence regularities. This design improves the structural reliability of pseudo-labels and alleviates error propagation caused by appearance-level ambiguity.

4.3. Instance-Level Geometric Prior for Structural Refinement

Although the category-level style prior improves category discrimination during pseudo-label generation, segmentation results for ACA components may still suffer from local omission, boundary discontinuity, and region fragmentation due to occlusion, scale variation, blurred boundaries, and structural damage. To further alleviate these local structural errors, we introduce an instance-level geometric prior π^{ip} for structural refinement.

Given an input image O , an external instance segmentation model first produces a set of candidate instance masks,

$$\mathcal{M}(O) = \{m_k\}_{k=1}^K, \quad (13)$$

where K denotes the number of candidate instances and m_k denotes the k -th instance mask. Based on these masks, we construct the instance-level geometric prior as

$$\pi^{\text{ip}} = g_I(\mathcal{M}(O)), \quad (14)$$

which provides local geometric cues such as boundary continuity, region connectivity, and shape completeness.

Unlike the category-level style prior, π^{ip} is used for local structural correction rather than category inference. It refines the predicted outputs by improving boundary alignment and structural completeness, thereby complementing the category-level prior from a geometric perspective.

Algorithm 1: Hypergraph-Guided Style Prior Inference

Input: labeled set $\mathcal{D}_l = \{(O_j, C_j, g_j)\}_{j=1}^J$,
unlabeled set $\mathcal{D}_u = \{(O_u, g_u)\}_{u=1}^U$,
class set $\mathcal{C}$
Output: style prior masks $\{\pi_u^{\text{sp}}\}_{u=1}^U$
Construct reference hyperedge library $\mathcal{E} = \{e_j = C_j\}_{j=1}^J$
from labeled images;
for $u = 1$ **to** U **do**
 Infer instance-level category evidence
 $\mathbf{z}_u \leftarrow \text{InferInstanceEvidence}(O_u)$;
 Retrieve candidate hyperedges
 $\mathcal{E}_{g_u} = \{e_j \in \mathcal{E} \mid g_j = g_u\}$;
 Compute matching scores
 $\Delta(e) = \sum_{c \in \mathcal{C}} \mathbf{z}_u(c) \mathbb{I}[c \in e], \quad \forall e \in \mathcal{E}_{g_u}$;
 Select best-matched hyperedge set
 $\mathcal{E}_u^* = \{e \in \mathcal{E}_{g_u} \mid \Delta(e) = \max_{e' \in \mathcal{E}_{g_u}} \Delta(e')\}$;
 Choose minimal-cardinality prototype hyperedge
 $e_u^* = \arg \min_{e \in \mathcal{E}_u^*} |e|$;
 Initialize candidate category set $P_u \leftarrow e_u^*$;
 Compute class support
 $f_u(c) = \sum_{e \in \mathcal{E}_u^*} \mathbb{I}[c \in e], \quad \forall c \in \mathcal{C}$;
 Perform support-based category completion
 $P_u^+ = P_u \cup \{c \mid \tau_l \leq f_u(c) \leq \tau_h\}$;
 Generate style prior mask
 $\pi_{u,c}^{\text{sp}} = \mathbb{I}[c \in P_u^+], \quad \forall c \in \mathcal{C}$;
return $\{\pi_u^{\text{sp}}\}_{u=1}^U$;

4.4. Training and Inference

After obtaining the category-level style prior π^{sp} and the instance-level geometric prior π^{ip} , the proposed framework applies them in the training and inference stages, respectively. During training, π^{sp} is used to generate more reliable pseudo-labels for unlabeled samples and to optimize the student network. During inference, π^{ip} is used to refine the predicted segmentation by improving local structural completeness and boundary consistency.

4.4.1. SP-guided Pseudo-label Generation and Optimization

For an unlabeled image $O_u \in \mathcal{D}_u$, the teacher network first extracts discriminative features X_u and outputs the initial segmentation logits:

$$Z_u = s_\theta(X_u). \quad (15)$$

According to the category-level style prior π_u^{sp} inferred in Section 4.2.2, we introduce it into the teacher output as a category-level structural bias term, yielding the adjusted logits

$$\hat{Z}_u(x, y, c) = Z_u(x, y, c) + \alpha \pi_{u,c}^{\text{sp}}, \quad (16)$$

where α is a modulation coefficient, and $\pi_{u,c}^{\text{sp}} \in \{0, 1\}$ indicates whether category c is supported by the current scene prior. Here, π_u^{sp} is broadcast along the spatial dimensions and added to each class channel of the teacher logits. Applying the softmax

operation to the adjusted logits gives the pseudo-label distribution:

$$P_s(Y_u(x, y) = c \mid O_u) = \frac{\exp(\hat{Z}_u(x, y, c))}{\sum_{c'=1}^{\mathcal{C}} \exp(\hat{Z}_u(x, y, c'))}. \quad (17)$$

Based on this distribution, we generate the pseudo-label $\hat{Y}_u$ for the unlabeled sample and use it as the supervisory signal for training the student network. For labeled samples $(O_l, Y_l) \in \mathcal{D}_l$, the standard supervised segmentation loss is used:

$$\mathcal{L}_{\text{sup}} = \frac{1}{|\mathcal{D}_l|} \sum_{(O_l, Y_l) \in \mathcal{D}_l} \ell(s_\theta(X_l), Y_l), \quad (18)$$

where $\ell(\cdot)$ denotes the supervised segmentation loss function. For unlabeled samples, the unsupervised loss based on pseudo-labels is defined as

$$\mathcal{L}_{\text{unsup}} = \frac{1}{|\mathcal{D}_u|} \sum_{O_u \in \mathcal{D}_u} \ell(s_\theta(X_u), \hat{Y}_u). \quad (19)$$

Accordingly, the overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_u \mathcal{L}_{\text{unsup}}, \quad (20)$$

where λ_u is the weighting coefficient for the unsupervised term.

By introducing π^{sp} into pseudo-label generation, the teacher network no longer relies solely on local high-response appearance cues, but instead adjusts the category distribution according to scene-level structural regularities. This helps reduce the accumulation of style-induced pseudo-label bias during self-training.

4.4.2. IP-based Prediction Refinement

During inference, given an input image O , the trained segmentation network first produces the initial prediction mask

$$\hat{Y}^{(0)}(x, y) = \arg \max_{c \in \mathcal{C}} \text{Softmax}(s_\theta(X))_{x,y,c}. \quad (21)$$

Based on the candidate instance masks $\mathcal{M}(O) = \{m_k\}_{k=1}^K$, the instance-level geometric prior introduced in Section 4.3 is used to refine the initial prediction. The final refined prediction is obtained by

$$\hat{Y} = \mathcal{T}(\hat{Y}^{(0)}, \pi^{\text{ip}}; \eta, r), \quad (22)$$

where $\mathcal{T}(\cdot)$ denotes the instance-guided refinement operator, η is the dominance threshold, and r is the normalized refinement coefficient for mask expansion. A detailed description of the refinement process is provided in Algorithm 2.

This refinement mainly addresses two types of structural errors: boundary inconsistency between predicted regions and instance contours, and insufficient local response that leads to holes, missing parts, or discontinuities within target objects. By introducing geometric constraints from instance masks, π^{ip} enables local structural correction without changing the overall category prediction.

Algorithm 2: Instance-Geometric-Prior-Guided Prediction Refinement

Input: initial prediction mask $\hat{Y}_u^{(0)}$ for image O_u ,
candidate instance masks $\mathcal{M}_u = \{m_{u,k}\}_{k=1}^{K_u}$,
dominance threshold η , normalized refinement
coefficient r , class set C
Output: refined prediction mask $\hat{Y}_u$
Initialize refined mask $\hat{Y}_u \leftarrow \hat{Y}_u^{(0)}$;
Convert r to the pixel-level dilation radius: $R_r \leftarrow \lfloor \kappa r \rfloor$;
for $k = 1$ **to** K_u **do**
 Compute class histogram
 $n_k(c) = |\{p \in m_{u,k} \mid \hat{Y}_u^{(0)}(p) = c\}|, \quad \forall c \in C$;
 Compute dominant class $c_k^* = \arg \max_{c \in C} n_k(c)$;
 Compute dominance ratio $\rho_k = \frac{n_k(c_k^*)}{|m_{u,k}|}$;
 if $\rho_k \geq \eta$ **then**
 Assign instance label $\tilde{c}_k \leftarrow c_k^*$;
 else
 Compute instance center $p_k^{\text{ctr}} \leftarrow \text{Center}(m_{u,k})$;
 Assign fallback label $\tilde{c}_k \leftarrow \hat{Y}_u^{(0)}(p_k^{\text{ctr}})$;
 Expand instance region $\tilde{m}_{u,k} \leftarrow \text{Dilate}(m_{u,k}, R_r)$;
 Refine local prediction by propagating the instance
 label: $\hat{Y}_u(p) \leftarrow \tilde{c}_k, \quad \forall p \in \tilde{m}_{u,k}$;
return $\hat{Y}_u$;

5. Experiments

This section first provides an overview of the datasets, experimental environment, and hyperparameter settings used for training and evaluation. We then present the overall performance of the proposed method, assessed through both quantitative metrics and qualitative visualizations. In addition, we verify the plug-and-play capability of the proposed framework across different baseline models. Finally, extensive ablation studies are conducted to highlight the contribution of each component in the framework.

5.1. Datasets and Evaluation Metric

ACA is a dataset for semantic segmentation in ancient Chinese architectural scenes, released by the National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences. It covers six representative religious architectural complexes and features pronounced intra-class variation and inter-class ambiguity, making it suitable for evaluating semi-supervised semantic segmentation in ACA scenes. ACA contains 17 foreground classes and 1 background class, namely architecture (AT), tsun-shou (TS), window (WD), stone elephant (SE), pillar (PI), stone step (SS), architrave (AR), stone tablet (SA), door (DR), ridge (RD), pagoda (PG), fence (FE), stone lion (SL), censer (CE), dougong (DG), stair (ST), plaque (PE), and background (BG). According to the dataset split, it contains 1308 training images and 145 validation images. For semi-supervised evaluation, we adopt four labeling protocols, namely 1/16, 1/8, 1/4, and 1/2, on the ACA training

set. Under each protocol, the corresponding proportion of training images is used as labeled data with pixel-wise annotations, while the remaining training images are treated as unlabeled samples. The validation set, as well as the labeled/unlabeled splits under each protocol, are kept fixed across all experiments.

PASCAL VOC2012 is a widely used benchmark for semantic segmentation, containing 20 foreground classes and 1 background class. According to the official split, it includes 1464 finely annotated training images and 1449 validation images. Following common semi-supervised semantic segmentation settings, we use the 1464 finely annotated training images as the candidate pool for labeled data and additionally introduce 9118 augmented images with coarse pixel annotations as unlabeled data. During training, these coarse annotations are not used; the augmented images are treated purely as unlabeled samples. For the labeling ratio, we follow standard protocols in prior work and sample labeled subsets from the 1464 training images under 1/16, 1/8, 1/4, 1/2, and Full (1464) settings. The remaining unsampled training images, together with the 9118 augmented samples, are used as unlabeled data. The validation set is kept unchanged under all protocols.

Evaluation metrics. In all experiments, mean Intersection-over-Union (mIoU) is adopted as the primary evaluation metric to assess overall segmentation performance across semantic categories. It reflects the average overlap between the predicted segmentation and the ground truth over all categories, and has been widely used in semantic segmentation tasks. Higher mIoU values indicate better segmentation performance.

5.2. Implementation details

To ensure fair comparison, all methods are implemented under the same hardware environment and a unified training protocol. All experiments are conducted on four NVIDIA Tesla V100 GPUs, with PyTorch used for training and evaluation. Unless otherwise specified, the input images are cropped to 322×322 , and the batch size is set to 8. The confidence threshold for pseudo labels in the unlabeled branch is fixed at 0.95. To mitigate overfitting under the limited ACA training data, we adopt the same augmentation protocol as UniMatch[7] for all compared methods. Specifically, weak augmentation consists of random resizing, random cropping, and horizontal flipping, while strong augmentation further includes color jittering, grayscale transformation, Gaussian blurring, and CutMix. Our main baseline is built on the UniMatch-v2 framework, where the segmentation model adopts a DPT architecture based on DINOv2, with DINOv2-Small as the encoder. During training, Complementary Dropout is enabled with a dropout probability of 0.5. The remaining comparison methods use ResNet-101 as the encoder and DeepLabv3+ as the decoder. For all methods, the learning rate, optimizer, and scheduling strategy follow the recommended settings in their original papers or official implementations, without additional task-specific tuning. All models are trained for 80 epochs to ensure a consistent training budget across comparisons. For the plug-and-play evaluation, the proposed modules are directly integrated into the corresponding baseline frameworks. The training resolution, batch size,

Table 2: Performance comparison with other state-of-the-art methods on the ACA dataset.

Methods	Publication	Backbone	Performances (%)			
			1/16 (82)	1/8 (164)	1/4 (327)	1/2 (654)
Supervised		ResNet-101	48.76	53.64	56.95	56.81
DAW[5]	NeurIPS 2023	ResNet-101	55.87	59.94	61.67	55.29
DDFP[21]	CVPR 2024	ResNet-101	48.70	59.53	58.11	58.07
Beyond-Pixels[23]	ECCV 2024	ResNet-101	52.17	59.78	58.50	55.16
CSL[17]	ICCV 2025	ResNet-101	54.81	61.31	61.63	57.81
UniMatch[7]	CVPR 2023	ResNet-101	52.72	59.35	59.42	54.74
UniMatch+SP+IP		ResNet-101	53.95	60.64	59.80	57.66
Δ			+(1.23)	+(1.29)	+(0.38)	+(2.92)
CorrMatch[6]	CVPR 2024	ResNet-101	45.26	57.87	50.26	52.63
CorrMatch+SP+IP		ResNet-101	45.59	59.12	50.69	53.52
Δ			+(0.33)	+(1.25)	+(0.43)	+(0.89)
RankMatch[20]	CVPR 2024	ResNet-101	50.10	58.35	58.07	53.23
RankMatch+SP+IP		ResNet-101	51.31	59.87	57.83	55.35
Δ			+(1.21)	+(1.52)	-(0.24)	+(2.12)
MCCL[22]	TMM 2025	ResNet-101	52.26	59.09	55.53	55.45
MCCL+SP+IP		ResNet-101	52.25	60.48	56.99	57.28
Δ			-(0.01)	+(1.39)	+(1.46)	+(1.83)
Supervised		DINOv2-S	54.86	61.07	61.23	61.67
UniMatch-v2[19]	TPAMI 2025	DINOv2-S	56.13	63.60	63.86	62.67
UniMatch-v2+SP+IP		DINOv2-S	58.26	64.30	65.43	65.16
Δ			+(2.13)	+(0.70)	+(1.57)	+(2.49)

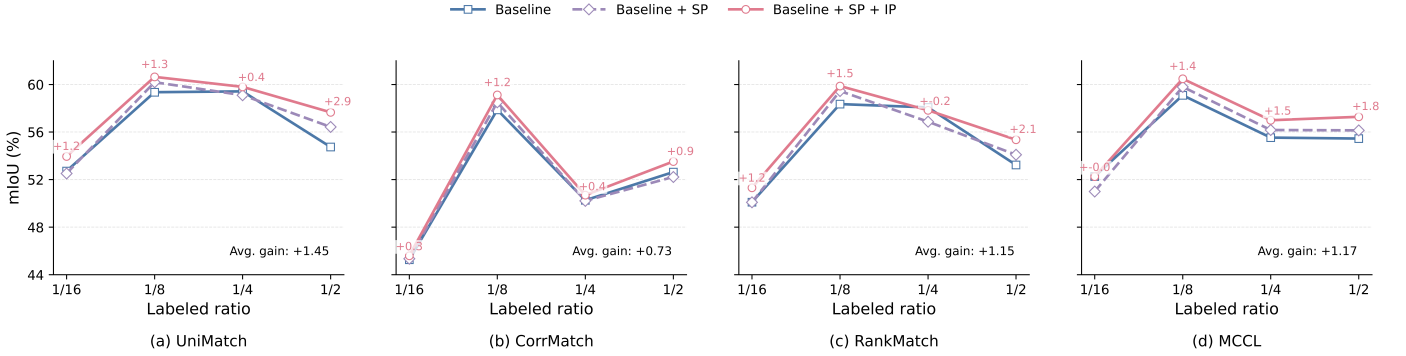


Figure 5: mIoU trends on ACA under different labeled ratios for four semi-supervised segmentation baselines with incremental SP and IP integration.

learning rate, and optimizer remain unchanged from the original settings, and only the parameters required by the proposed modules are additionally introduced.

5.3. Comparison with State-of-the-Art Methods

Results on the ACA dataset. Table 2 compares the proposed method with existing semi-supervised semantic segmentation approaches on ACA. After incorporating SP and IP, all baselines show consistent mIoU improvements across different labeling ratios. As illustrated in Fig. 5, the performance curves of UniMatch, CorrMatch, RankMatch, and MCCL are consistently shifted upward, with more pronounced gains under low-label settings. For instance, with ResNet-101, UniMatch improves by 1.23 under 1/16 supervision and by 2.92 under 1/2 supervision. With DINOv2-S, UniMatch-v2 achieves further gains of 2.13 and 2.49 under the same settings. These results suggest that the proposed priors provide useful semantic and structural guidance under semi-supervised learning.

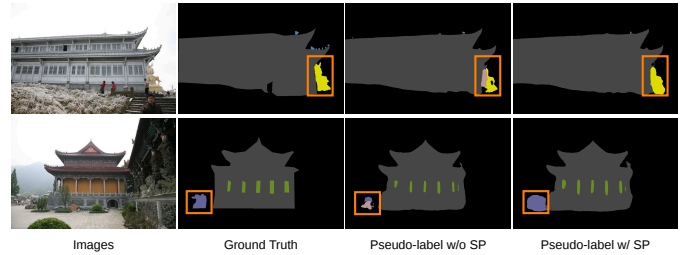


Figure 6: Visualization of UniMatch-v2 teacher pseudo-labels before and after introducing SP. Orange boxes highlight regions where SP suppresses implausible stone-lion responses in pagoda and stone-elephant regions.

Table 3 presents the category-level results at the 1/8 label ratio. SP is particularly effective for stone-component categories with similar appearance and material properties, which are more susceptible to inter-class confusion. For example, the

Table 3: Per-class IoU (%) comparison on ACA.

Methods	BG	AT	ST	WD	PI	TS	DR	SS	AR	PE	DG	RD	PG	SL	SE	FE	CE	SA	mIoU
DAW[5]	84.54	81.00	70.43	60.61	50.54	30.45	37.66	39.15	44.72	63.75	55.37	40.67	87.39	81.03	71.62	59.26	75.30	45.41	59.94
DDFP[21]	85.46	80.54	69.44	49.47	43.35	32.60	43.01	46.15	41.86	67.92	55.04	42.13	82.45	70.92	72.39	59.58	75.09	54.07	59.53
CSL[17]	84.23	81.16	70.43	60.92	45.40	32.45	54.47	41.01	43.27	61.97	57.23	42.62	88.98	83.39	72.37	64.44	73.20	46.13	61.31
Beyond-Pixels[23]	84.25	81.00	67.63	59.50	45.89	29.19	38.42	39.98	48.64	63.48	59.80	38.39	87.84	80.38	65.99	60.02	70.69	54.97	59.78
UniMatch[7]	84.14	80.83	67.87	58.74	44.90	24.92	39.53	40.21	50.74	62.91	58.16	37.12	86.78	80.17	62.80	58.28	69.12	61.04	59.35
UniMatch+SP+IP	84.47	80.67	68.80	59.94	50.58	30.19	39.80	42.26	47.99	65.58	60.41	34.56	87.97	80.11	73.90	51.53	74.37	58.36	60.64
CorrMatch[6]	84.40	80.74	64.73	55.82	38.78	20.67	35.84	44.86	41.73	62.65	54.94	40.00	87.32	80.91	69.75	56.45	70.96	51.15	57.87
CorrMatch+SP+IP	84.81	80.47	68.92	55.92	41.28	22.87	36.54	55.67	42.21	63.05	52.90	42.61	86.89	80.62	69.88	56.49	73.43	49.63	59.12
RankMatch[20]	83.80	79.75	61.13	61.44	44.46	28.47	38.17	33.05	40.78	62.24	57.21	37.43	88.19	79.81	72.09	49.69	71.09	61.40	58.35
RankMatch+SP+IP	84.43	80.78	65.50	62.20	44.61	32.10	38.97	44.16	40.24	63.80	54.39	36.45	87.55	83.14	72.92	61.40	69.19	55.86	59.87
MCCL[22]	84.29	80.78	70.57	57.52	45.16	28.25	36.35	41.82	44.81	60.43	58.09	34.93	87.45	78.53	65.29	56.67	71.82	60.95	59.09
MCCL+SP+IP	84.06	80.13	71.61	60.14	52.22	29.09	36.99	49.53	44.03	61.89	56.84	34.13	87.78	80.68	73.20	54.58	72.80	58.88	60.48
UniMatch-v2[19]	85.83	82.19	69.20	58.77	69.15	30.16	35.94	47.26	40.13	68.95	62.43	35.47	86.36	87.92	74.35	62.97	72.75	74.89	63.60
UniMatch-v2+SP+IP	85.91	82.24	68.62	63.22	67.22	31.67	38.32	50.16	39.61	69.43	61.59	39.06	85.16	88.84	77.51	62.36	73.51	72.98	64.30

Table 4: Comparison on PASCAL VOC2012 under Different Labeling Ratios with Instance-level Geometric Prior (IP).

Methods	Publication	Backbone	Performances (%)				
			1/16 (92)	1/8 (183)	1/4 (366)	1/2 (732)	Full (1464)
Supervised		ResNet-101	45.1	55.3	64.8	69.7	73.5
DAW[5]	NeurIPS 2023	ResNet-101	74.8	77.4	79.5	80.6	81.5
DDFP[21]	CVPR 2024	ResNet-101	75.0	78.0	79.5	81.2	82.0
CSL[17]	ICCV 2025	ResNet-101	76.8	79.6	80.3	80.9	82.3
UniMatch[7]	CVPR 2023	ResNet-101	75.2	77.2	78.8	79.9	81.2
CorrMatch[6]	CVPR 2024	ResNet-101	76.4	78.5	79.4	80.6	81.8
Beyond-Pixels[23]	ECCV 2024	ResNet-101	77.3	78.6	79.8	80.8	81.7
RankMatch[20]	CVPR 2024	ResNet-101	75.5	77.6	79.8	80.7	82.2
Supervised		DINOv2-S	67.0	75.6	81.8	83.7	85.6
UniMatch-v2[19]	TPAMI 2025	DINOv2-S	78.96	85.50	85.83	86.62	87.74
UniMatch-v2+IP		DINOv2-S	79.13	85.53	85.86	86.66	87.83

performance of stone elephant (SE) increases from 62.80 to 73.90 (+11.10) in UniMatch and from 65.29 to 73.20 (+7.91) in MCCL, indicating improved discriminability among visually similar categories. By contrast, IP is more effective in mitigating detail loss and structural fragmentation caused by large intra-class variation. Specifically, pillar (PI) improves by 5.68 in UniMatch, window (WD) improves by 4.45 in UniMatch-v2, and PI further improves by 7.06 in MCCL. These findings suggest that SP primarily enhances discriminability for confusing categories, whereas IP mainly improves detail preservation and shape consistency for structurally sensitive categories. Their combination is therefore complementary and yields stable performance gains across different label-ratio settings.

Results on the PASCAL VOC2012 dataset. Table 4 reports the quantitative results on the original PASCAL VOC2012 dataset under different label-ratio settings. “Supervised” denotes training with only the labeled data at the corresponding ratio, without using unlabeled samples. UniMatch-v2 achieves strong performance on this general-purpose dataset, indicating its ability to effectively exploit unlabeled data and learn stable semantic representations. However, VOC is an object-centric dataset with sparse category co-occurrence and limited scene-level compositional structure, making it unsuitable for reliable modeling of scene-level style priors. Therefore, we only evaluate the transferability of the IP module and do not report results

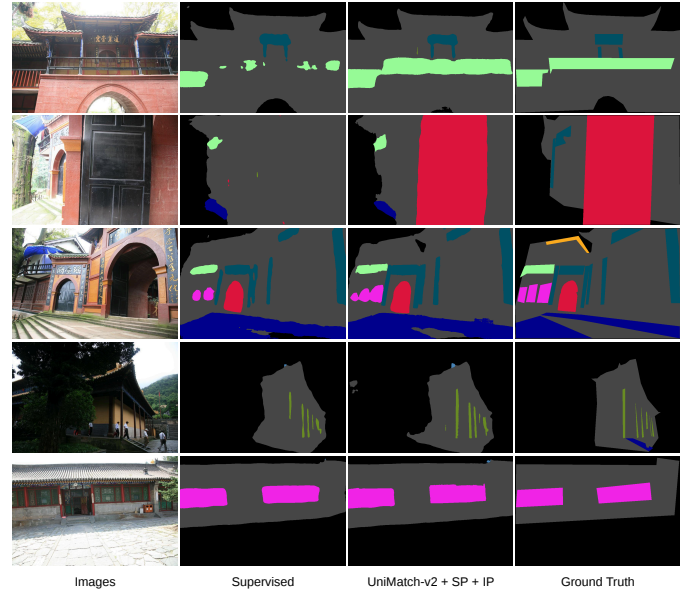


Figure 7: Comparison of the proposed method with the supervised-only baseline.

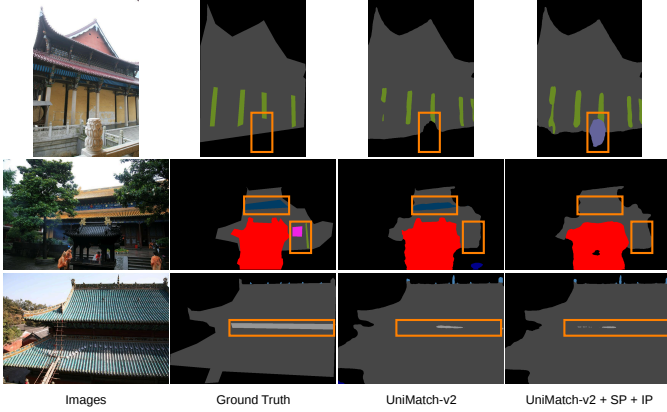


Figure 8: Representative failure cases of C-PGS on the ACA dataset. Orange boxes highlight typical remaining segmentation errors.

for the full SP+IP setting.

The VOC experiment is intended to examine the cross-dataset applicability of the instance-level geometric refinement strategy, rather than the effectiveness of the complete C-PGS framework in object-centric scenes. Under this setting, IP is integrated into UniMatch-v2 in a plug-and-play manner. As shown in Table 4, the improvement on VOC2012 is marginal. This is likely because the dominant errors of UniMatch-v2 on VOC are mainly caused by local semantic misclassification rather than boundary instability or missing geometric structures. Since geometric priors primarily enhance boundary and shape consistency, their effect is naturally limited when errors are dominated by semantic ambiguity.

Visual comparison of structural segmentation. Fig. 6 provides direct visual evidence of the effect of SP on pseudo-label generation in UniMatch-v2. Compared with the original teacher pseudo-labels, the SP-guided pseudo-labels suppress implausible stone-lion responses in the pagoda and stone-elephant regions, suggesting that the style prior improves pseudo-label reliability for visually similar categories. The qualitative comparison in Fig. 7 shows that, compared with the baseline trained only on labeled samples, the proposed method produces more complete responses, less category confusion, and clearer boundaries for key architectural components. In Fig. 9, different baselines and their C-PGS-enhanced variants are compared on the same input images, where colored boxes and enlarged local regions highlight representative local differences. Overall, the proposed priors lead to cleaner target regions, clearer boundaries, and fewer fragmented predictions than competing methods. The second and third rows mainly illustrate stone-component scenes, where target components are easily confused with visually similar stone-like surroundings; the enlarged regions show that C-PGS suppresses noisy fragments and produces cleaner predictions, indicating that SP alleviates appearance-induced category confusion. The first and fourth rows focus on fine-grained structures, such as windows, fences, and pillar-like components, where the enhanced results exhibit more continuous shapes and clearer boundaries, demonstrating that IP improves structural completeness and boundary

Table 5: Sensitivity analysis of key hyperparameters on ACA using UniMatch-v2.

1/4 labeled		1/2 labeled					
α		τ		η		r	
value	mIoU	value	mIoU	value	mIoU	value	mIoU
0.4	62.18	0	61.56	0.5	65.11	0	64.61
0.5	64.17	0.7	62.14	0.6	65.16	0.4	65.14
0.6	65.20	0.8	60.90	0.7	65.04	0.5	65.15
0.7	62.14	0.9	63.10	0.8	65.00	0.6	65.16
0.8	63.36	0.95	64.48	0.9	64.55	0.7	65.16
–	–	0.98	63.92	–	–	0.8	65.16

Table 6: Effect of BGK on ACA under different labeled ratios (UniMatch-v2).

Methods	Performances (%)			
	1/16 (82)	1/8 (164)	1/4 (327)	1/2 (654)
w/o BGK	56.13	63.60	63.86	62.67
w BGK	56.92	62.87	64.82	62.98

consistency.

Failure case analysis. As shown in Fig. 8, the remaining failure cases of C-PGS mainly arise from three aspects. First, highly similar appearances may still cause category confusion or missed predictions. For instance, the stone post on the railing in the first row is misclassified as a stone elephant. Similarly, in the second row, the windows and columns are partially missed due to their high visual similarity to the surrounding building body. Second, if the initial prediction does not activate a target region, IP cannot effectively refine it, as shown by the missing architrave region in the second row. Third, the refinement effect is limited when SAM2 fails to provide reliable instance masks, such as the dougong region in the third row. These cases suggest that robust recognition under severe appearance similarity and incomplete instance guidance remains an important direction for future work.

5.4. Ablation study

Parameter analysis. We conduct an ablation study on several key hyperparameters in the UniMatch-v2 framework, and summarize the results in Table 5. Specifically, α controls the initial strength of the style prior, τ is the reliable-pixel threshold, and η and r are the main hyperparameters of the instance-level geometric prior (IP). A value of 0 means that the corresponding mechanism is disabled.

For the style-prior strength α , experiments under the 1/4 labeling protocol show that a moderate prior strength leads to the best performance. Specifically, the model achieves the highest mIoU of 65.20 when $\alpha = 0.6$, outperforming both smaller values such as $\alpha = 0.4$ (62.18) and larger ones such as $\alpha = 0.7$ (62.14). This suggests that an appropriate initial prior strength can more effectively suppress spurious responses caused by style similarity during early training, whereas an overly small value provides insufficient constraint and an overly large one may introduce unnecessary bias. For the reliable-pixel threshold τ , results under the 1/2 labeling protocol in-

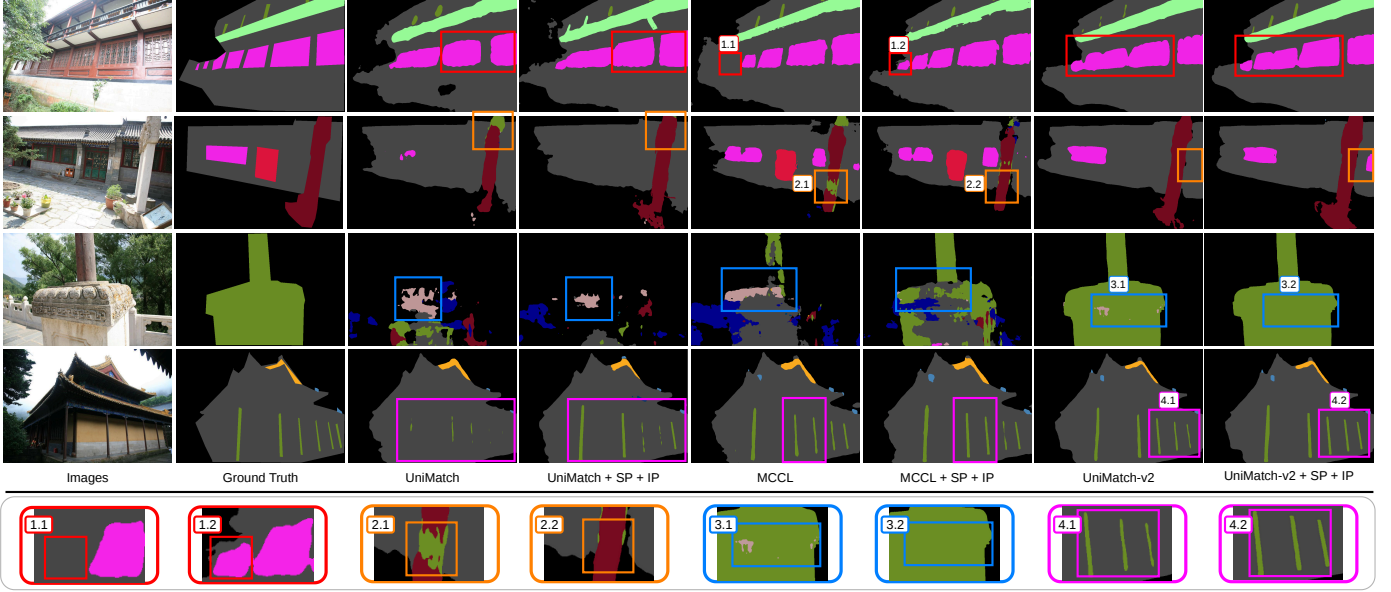


Figure 9: Qualitative comparison on the ACA dataset. Each row shows the same input image across different methods, with colored boxes and enlarged local regions highlighting representative local differences.

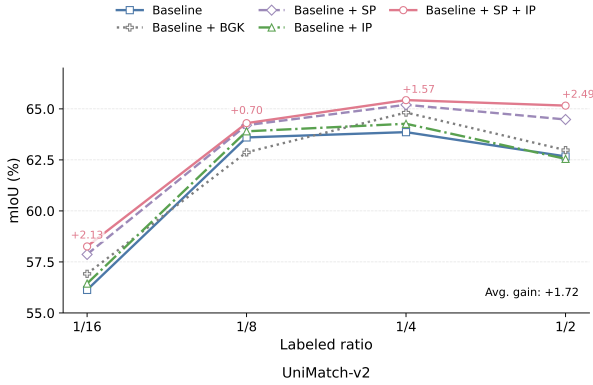


Figure 10: Component-wise ablation on UniMatch-v2 under different labeled ratios on ACA.

indicate that pseudo-label filtering is beneficial. When filtering is disabled ($\tau = 0$), the model obtains 61.56 mIoU. After introducing reliable-pixel selection, the performance improves consistently and reaches the best result of 64.48 at $\tau = 0.95$, indicating that filtering unreliable regions effectively reduces noise and allows subsequent updates to focus on high-confidence predictions. For the IP-related hyperparameters, both η and r show relatively stable behavior over a broad range under the 1/2 labeling protocol. In particular, η achieves the best result of 65.16 mIoU at $\eta = 0.6$, while remaining highly stable for nearby values, which suggests that the intra-instance decision threshold is fairly robust. For the normalized refinement coefficient r , increasing it from 0 yields consistent improvement: the performance rises from 64.61 at $r = 0$ and reaches 65.16 when $r \geq 0.6$, after which it remains stable. This indicates that moderate instance-guided expansion helps improve structural continuity and local completeness, while excessive expansion pro-

Table 7: Ablation and complexity analysis of the proposed SP and IP modules on ACA under the 1/4 labeled setting (UniMatch-v2).

Baseline	SP	IP	mIoU	Aux. Params	Train Time	Infer Time	Memory
✓			63.86	—	0.88 s	0.04 s	10.35 G
✓	✓		65.20	0.101M	0.87 s	0.04 s	10.41 G
✓		✓	64.27	0	0.88 s	3.63 s	10.35 G
✓	✓	✓	65.43	0.101M	0.87 s	3.44 s	10.41 G

vides little further benefit.

Effect of building-group knowledge. Table 6 evaluates the effect of building-group knowledge (BGK) on UniMatch-v2. In this work, BGK denotes the architectural-complex information associated with each image and is used as a coarse grouping cue for style-hypergraph construction and style-prior retrieval, rather than as pixel-level supervision or direct category annotation. This setting is designed to examine whether architectural-complex metadata itself brings substantial gains, and to distinguish its effect from that of the proposed style-prior modeling.

As shown in Table 6, introducing BGK leads to only limited and inconsistent performance changes. The mIoU improves from 56.13 to 56.92 under 1/16 supervision, from 63.86 to 64.82 under 1/4, and from 62.67 to 62.98 under 1/2, while slightly decreasing from 63.60 to 62.87 under 1/8. This trend is also reflected in Fig. 10, where the BGK curve remains close to the baseline and does not show a consistent upward shift across labeling ratios. These results suggest that BGK can provide auxiliary scene-level information in some cases, but it is not the dominant factor behind the performance gain. The main improvement of C-PGS comes from the proposed style-prior modeling, which captures structural category regularities and uses them to guide pseudo-label generation.

Effect of each component. Table 7 reports the contribution

and computational overhead of SP and IP under the UniMatch-v2 framework with 1/4 labeled data. Aux. Params, Train Time, Infer Time, and Memory denote task-specific auxiliary parameters, average training time per iteration, average inference time per image, and peak GPU memory, respectively. The baseline achieves 63.86 mIoU, while adding SP improves the performance to 65.20 (+1.34), showing that SP is the main contributor by alleviating inter-class confusion and style-induced pseudo-label bias propagation. SP introduces only 0.101M auxiliary parameters and barely affects inference time, since it is mainly used during pseudo-label generation. Adding IP increases the mIoU to 64.27 (+0.41), indicating that instance-level geometric constraints help improve shape consistency and detail preservation, with its overhead mainly reflected in inference time due to SAM2-based mask generation and refinement. Using SP and IP together achieves the best result of 65.43 mIoU (+1.57), with only a slight increase in memory usage. These results indicate that SP and IP are complementary and can be independently plugged into semi-supervised segmentation frameworks without modifying the segmentation backbone.

6. Conclusion

In this work, we studied semi-supervised semantic segmentation for ACA images with a focus on style-induced pseudo-label bias propagation. We found that the difficulty of this task comes from limited supervision together with models reliance on unstable appearance cues under large intra-class variation and high inter-class similarity. To address this issue, we introduced C-PGS, which combines a category-level style prior to improve pseudo-label reliability with an instance-level geometric prior to enhance structural completeness and boundary consistency during inference. Results on multiple mainstream semi-supervised semantic segmentation baselines show that the proposed method consistently improves performance across different labeling ratios. Overall, these results highlight the value of combining structural style knowledge with geometric cues for improving segmentation quality in ACA scenes.

Future work will focus on further strengthening the discrimination of visually similar fine-grained components and developing more expressive structural priors by incorporating relative spatial positions, topological relations, and richer scene organization cues. We also plan to explore how the proposed framework can be extended to a broader range of heritage or structured-scene segmentation tasks beyond the current ACA setting.

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